

Extra Credit Homework

Deadline: April 24, 2026 11:59pm ET

Instructions

- This is an *extra credit* homework. Answer as many questions as you would like. All correct answers will be counted as extra credit.

Problems

- (10 points) In this problem we will see why a natural idea for constructing an IND-CCA-2 secure public key encryption scheme does not work. Let $(\text{KGen}, \text{Enc}, \text{Dec})$ be an IND-CPA secure public key encryption scheme, and let $(\text{DSKeyGen}, \text{Sign}, \text{Ver})$ be a strong UF-CMA secure digital signature scheme. We define a new public key encryption scheme $(\text{KGen}', \text{Enc}', \text{Dec}')$ below.

$\text{KGen}'(1^\lambda)$	$\text{Enc}'(\text{pk}, m)$	$\text{Dec}'(\text{sk}, (\text{pk}_{\text{sig}}, \text{ct}, \sigma))$
1 : return $\text{KGen}(1^\lambda)$	1 : $(\text{sk}_{\text{sig}}, \text{pk}_{\text{sig}}) \leftarrow \text{DSKeyGen}(1^\lambda)$	1 : if $\text{Ver}(\text{pk}_{\text{sig}}, \text{ct}, \sigma) \neq 1$ then return $\perp$
	2 : $\text{ct} \leftarrow \text{Enc}(\text{pk}, m)$	2 : else return $\text{Dec}(\text{sk}, \text{ct})$
	3 : $\sigma \leftarrow \text{Sign}(\text{sk}_{\text{sig}}, \text{ct})$	
	4 : return $(\text{pk}_{\text{sig}}, \text{ct}, \sigma)$	

Prove that $(\text{KGen}', \text{Enc}', \text{Dec}')$ does *not* satisfy IND-CCA-2 security.

- (10 points) We say that a public key encryption scheme $(\text{KGen}, \text{Enc}, \text{Dec})$ is *additively homomorphic* if for all $m_0, m_1 \in \mathcal{M}$:

$$\Pr \left[\begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{KGen}(1^\lambda) \\ \text{Dec}(\text{sk}, \text{ct}_0 + \text{ct}_1) = m_0 + m_1 : \begin{array}{l} \text{ct}_0 \leftarrow \text{Enc}(\text{pk}, m_0) \\ \text{ct}_1 \leftarrow \text{Enc}(\text{pk}, m_1) \end{array} \end{array} \right] = 1$$

Let $\Pi = (\text{KGen}, \text{Enc}, \text{Dec})$ be an additively homomorphic IND-CPA secure encryption scheme with message space $\mathcal{M} = \mathbb{Z}_p$. Prove that Π does *not* satisfy IND-CCA-2 security.

- (10 points) Let $(\text{KGen}, \text{Enc}, \text{Dec})$ be an encryption scheme. We can construct a new encryption scheme $(\text{KGen}', \text{Enc}', \text{Dec}')$ for ℓ -bit messages as follows:

$\text{KGen}'(1^\lambda)$	$\text{Enc}'(k, m; r)$	$\text{Dec}'(k, \text{ct})$
1 : return $\text{KGen}(1^\lambda)$	1 : return $\text{Enc}(k, m r; r)$	1 : return $\text{Dec}(k, \text{ct})[: \ell]$

Recall that the notation $\text{Enc}(k, m; r)$ means explicitly defining the randomness used in Enc . You may assume that $r \in \{0, 1\}^\lambda$, and therefore that executing $\text{Enc}(k, m)$ is equivalent to first sampling $r \leftarrow \{0, 1\}^\lambda$ and then running $\text{Enc}(k, m; r)$.

We will prove that the above construction is not secure.

- (a) Let $(\text{KGen}^*, \text{Enc}^*, \text{Dec}^*)$ be an IND-CPA secure encryption scheme. Construct an encryption scheme $(\text{KGen}, \text{Enc}, \text{Dec})$, and briefly justify why it is correct and satisfies IND-CPA security (you do not need to write a formal proof).
- (b) Prove that when $(\text{KGen}', \text{Enc}', \text{Dec}')$ is instantiated with your scheme the resulting scheme does *not* satisfy IND-CPA security.
4. (20 points) Suppose a family of functions $H = \{h_i : \{0, 1\}^{2\lambda} \rightarrow \{0, 1\}^\lambda\}$ has the following property: for all h_i , it is the case for all $x, y \in \{0, 1\}^{2\lambda}$ that if $x \leq y$ then $h_i(x) \leq h_i(y)$. Prove that H is *not* a collision resistant hash function family. Your proof should give an explicit collision-finding attack against H .
5. (20 points) Suppose that Alice and Bob hold correlated inputs of the following form: Alice has (r_0, r_1) , where each $r_i \leftarrow_{\$} \{0, 1\}$ and Bob has (c, r_c) , where $c \leftarrow_{\$} \{0, 1\}$.

Further suppose that at a later point, Alice and Bob wish to securely compute 1-out-of-2 OT with inputs (x_0, x_1) and b respectively (assume that x_0 and x_1 are also single bits). Show how Alice and Bob can use their correlated inputs to perform this task without using any cryptographic assumptions. That is, design a protocol for 1-out-of-2 OT that achieves *unconditional* security against semi-honest adversaries. Argue correctness and security of your protocol. (You don't need to give a full formal proof.)

Hint: Recall that one-time pads do not require any cryptographic assumptions.

6. (10 points) Let Alice and Bob be two parties with inputs $z \in \mathbb{Z}_p$ and $b \in \mathbb{Z}_p$, respectively. They wish to check whether their inputs are equal, i.e. whether $a = b$. They want to do this while making sure they do not learn any other information about the other party's input. In other words, if $a \neq b$, then Alice should not learn anything about b and Bob should not learn anything about a .

Let $\mathbb{G}$ be a cyclic group of prime order p with generator g . Alice and Bob run the following protocol:

- Alice samples a random value $r \leftarrow_{\$} \mathbb{Z}_q$. She then computes $X = g^r$ and $Y = g^{ar}$. She sends (X, Y) to Bob.
- Bob computes X^b . He outputs 1 if $X^b = Y$, and 0 otherwise.

Explain why this protocol is not secure against a semi-honest Bob.

7. (20 points) Let Alice and Bob have inputs a and b , respectively. They want to securely send $(a + b)$ to a third-party Carol. Devise a protocol where Alice and Bob are only allowed to send **at most one message to each other** and **at most one message each to Carol**. Your protocol should satisfy all of the following security properties:

- *Security against semi-honest Alice:* Alice should not learn b .
- *Security against semi-honest Bob:* Bob should not learn a .
- *Security against semi-honest Carol:* Carol should not learn a or b .

Argue that your protocol indeed satisfies all three security conditions, and gives the correct output to Carol. (You do not need to give a formal proof.)